

Semantic Standardization of Unstructured Categorical Data Using Local LLMs and RAG: A Case Study on Industrial Tool Names

AI-Robot

Hongchul Shin

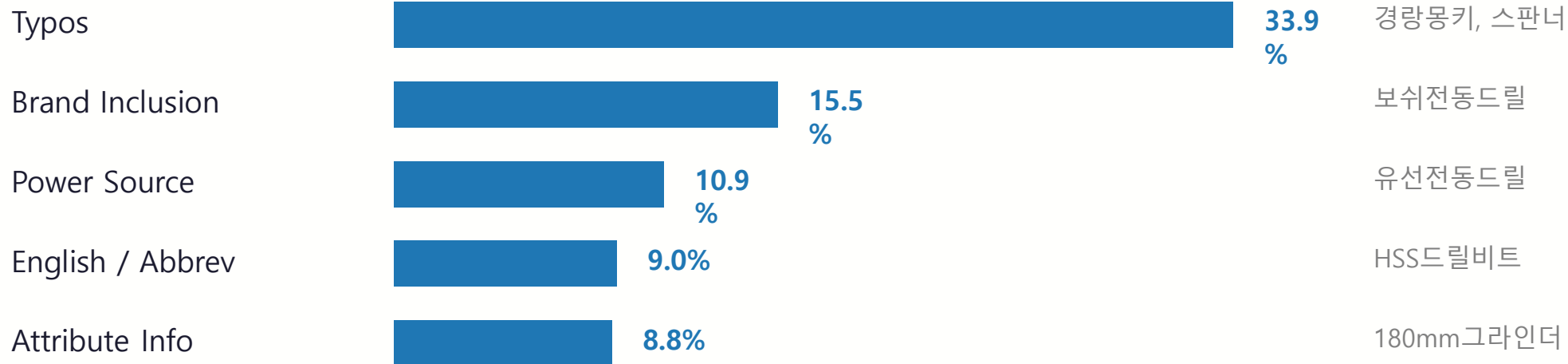
Research Highlight

1. Embedding selection comes first over modal size
2. The language of RAG can go beyond LLM's own limits.
3. RAG is essential for the success rate of refinement of specific domain data.



Why Is This Hard? – Error Patterns

78% of errors fall into 5 Systematic patterns – Non fixable by simple rules



Top 5 Combined: 78.1%

⚠ Japanese-derived terms (4.6%): 기리 / 뺨치 / 함마

Phonetic discontinuity → automation impossible without a thesaurus

The Problem: A Public Database Full of Aliases

Same tool –dozens of names – one broken database

드릴비트세트

기리셋트

giri set

기리셀트

drill bit SET

해머드릴

함마드릴

함마 드릴

HAMMER DRILL

함마드릴기

파이프렌치

PIPE렌치

파이프 렌치

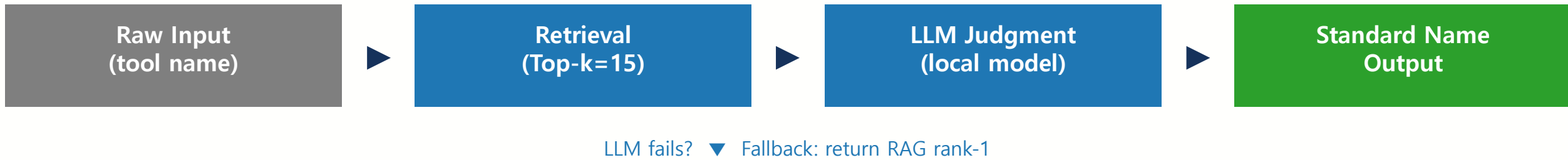
pipe wrench

Non-standard entries: 3,352 | Standard names: 160 | ~21 aliases per standard name

⚠ Broken standardization → broken inventory, broken search, broken statistics.

Our Approach: RAG Pipeline

Three targeted upgrades over the baseline



Component	Baseline (gray)	RAG (Improved)
Embedding	MiniLM (384-dim)	BGE-m3-ko (1024-dim) ★
Retrieval	FAISS only, k=5	BM25 40% + FAISS 60%, k=15
Safety net	—	Fallback (rank-1 auto-return)

The Verdict: RAG Transforms Accuracy

RAG is not optional

Gemma4:e4b



LLM-only 27.4%

+61.1pp



RAG 88.47% ★ BEST

Gemma3:4b



LLM-only 23.0%

+61.5pp



RAG 84.48%

Exaone3.5:7.8b



LLM-only 9.4%

+69.8pp



RAG 79.16%

DeepSeek Paradox: Korean generation 0% (LLM-only)



78.05% with RAG



RAG compensates for absent language capability.

KIST School, UST 2026 Academic Workshop

* 451 test samples · Exact Match Accuracy

What Drives Performance? – Embedding Matters Most

Same LLM. Different pipeline.

Model	Baseline (MiniLM + FAISS k=5)	RAG v2 Improved
Granite4.1:8b	55.21%	82.48% (+27.3pp)
Deepseek-r1:8b	55.21%	77.83% (+22.6pp)

Performance gain decomposition (estimated):

	Embedding swap (MiniLM → BGE-m3-ko)	~15–20pp	← DOMINANT
	Candidate pool (k=5 → k=15)	~3–5pp	
	BM25 hybrid (FAISS → BM25+FAISS)	~3–5pp	

Embedding selection dominates pipeline performance.
Choose your embedding before tuning your LLM.

The Problem: A Public Database Full of Aliases

Same LLM. Different

① RAG is essential

+61–70pp accuracy gain over LLM-only

② Small model wins

4B Gemma4:e4b → 88.47% exact match

③ Embedding first

BGE-m3-ko swap alone → +22–27pp

⚠ Remaining challenge: Japanese-derived terms → 47.1% | Synonym dictionary integration needed

"Privacy-safe, cost-effective, and fully deployable on-premise."